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System Programming

10Week [13 April - 19 April]

SOC2040 SP QUIZ4 SPRING 2024

Started on	2024-04-26 22:35
State	Finished
Completed on	2024-04-27 23:58
Time taken	1 day 1 hour
Marks	1265.00/1615.00
Grade	7.83 out of 10.00 (78%)

Question 1

Correct

Mark 10.00 out of 10.00

RDRAM stands for

Answer: Rambus DRAM



The correct answer is: Rambus DRAM

Question 2

Correct

Mark 5.00 out of 5.00

Which one of the following cache mapping approach requires no replacement algorithm.

Select one:

- ☐ 1. 4-way Set associative Cache
- ☐ 2. 8-way Set Associative Cache
- ☐ 3. Fully Associative Cache
- ☒ 4. Direct Mapped Cache



The correct answer is: Direct Mapped Cache

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Question 3

Correct

Mark 15.00 out of 15.00

Given a Seagate Cheetah 15K.4 Hard disk drive containing 4 platters having a zone map in the table shown below.

Determine the number of spare cylinders in the Zone 4.

Zone Number	Sectors per track	Cylinders per zone	Logical Blocks per zone
0	864	3201	22,076,928
1	844	3200	21,559,136
2	816	3400	22,149,504
3	806	3100	19,943,664
4	795	3100	19,671,480
5	768	3400	20,852,736
6	760	3450	21,159,936
7	725	3650	21,135,200
8	704	3700	20,804,608
9	672	3700	19,858,944
10	640	3700	18,913,280
11	603	3700	17,819,856
12	576	3707	17,054,208
13	528	3060	12,900,096

Select one:

- ☐ 1. 9
- ☐ 2. 6
- ☒ 3. 7 ✓
- ☐ 4. 8

The correct answer is: 7

Question 4

Correct

Mark 10.00 out of 10.00

..... refers to the tendency of execution to involve a number of memory locations that are clustered.

Answer: Spatial locality ✓

The correct answer is: Spatial locality

Question 5

Correct

Mark 5.00 out of 5.00

In Cache memory, each Block or line is a collection of

Answer: words ✓

The correct answer is: words

Question **6**

Correct

Mark 5.00 out of 5.00

Select the volatile memory device from the list below :

Select one:

- ☐ 1. EPROM
- ☐ 2. FLASH MEMORY
- ☐ 3. EEPROM
- ☒ 4. SRAM ✓

The correct answer is: SRAM

Question **7**

Correct

Mark 5.00 out of 5.00

..... library interpositioning requires access to a program's source files.

Select one:

- ☒ 1. Compile time ✓
- ☐ 2. Not a single
- ☐ 3. Run-time
- ☐ 4. Link-time

The correct answer is: Compile time

Question **8**

Correct

Mark 5.00 out of 5.00

In a typical ELF(Executable and Linkable Format) relocatable object file, which section/part of the file contains Symbol table with information about functions and global variables that are defined and referenced in the program.

Select one:

- ☐ 1. .debug section
- ☐ 2. .strtab section
- ☒ 3. .symtab section ✓
- ☐ 4. Section header table
- ☐ 5. .rel.data section
- ☐ 6. .rel.text section
- ☐ 7. .line section

The correct answer is: .symtab section

Question **9**

Correct

Mark 5.00 out of 5.00

An is a collection of concatenated relocatable object files, with a header that describes the size and location of each member object file.

Answer: ✓

The correct answer is: archive

Question **10**
Correct
Mark 10.00 out of 10.00

..... refers to the tendency for a processor to access memory locations that have been used recently.

Answer: Temporal locality



The correct answer is: Temporal locality

Question **11**
Correct
Mark 10.00 out of 10.00

Linking performed at run time is known as

Answer: Dynamic linking



The correct answer is: Dynamic linking

Question **12**
Correct
Mark 5.00 out of 5.00

..... library inter-positioning requires access to its relocatable object files

Select one:

- ☐ 1. Compile time
- ☒ 2. Link-time ✓
- ☐ 3. Run-time
- ☐ 4. Not a single

The correct answer is: Link-time

Question 13

Correct

Mark 15.00 out of 15.00

Suppose that a 1 MB file consisting of 512-byte logical blocks is stored on a disk drive with the following characteristics:

Parameter	Value
-----	-----
Rotational rate	10,000 RPM
Tavg seek	5 ms
Average # sectors/track	1024
Surfaces	4
Sector size	512 bytes

For the best case below, suppose that a program reads the logical blocks of the file sequentially, one after the other, and that the time to position the head over the first block is Tavg seek + Tavg rotation.

Best case: Estimate the optimal time (in ms) required to read the file given the best possible mapping of logical blocks to disk sectors (i.e., sequential).

Select one:

- ☐ 1. 12ms
- ☒ 2. 20ms ✓
- ☐ 3. 8ms
- ☐ 4. 10ms

The correct answer is: 20ms

Question 14

Correct

Mark 15.00 out of 15.00

Suppose that a 16 MB file consisting of 512-byte logical blocks is stored on a disk drive with the following characteristics:

Parameter	Value
-----	-----
Rotational rate	10,000 RPM
Tavg seek	5 ms
Average # sectors/track	1024
Platters	16
Sector size	512 bytes

For the random case below, suppose that a program reads the logical blocks of the file sequentially, one after the other, and that the time to position the head over the first block is Tavg seek + Tavg rotation.

Random case: Estimate the time (in ms) required to read the file if blocks are mapped randomly to disk sectors.

Select one:

- ☐ 1. 163.840secs
- ☐ 2. 32.768secs
- ☐ 3. 131.72secs
- ☒ 4. 262.144secs ✓

The correct answer is: 262.144secs

Question **15**

Correct

Mark 5.00 out of 5.00

..... is used to construct Main memory

Select one:

- ☐ 1. EEPROM
- ☒ 2. DRAM ✓
- ☐ 3. EPROM
- ☐ 4. SRAM

The correct answer is: DRAM

Question **16**

Correct

Mark 5.00 out of 5.00

In a typical ELF(Executable and Linkable Format) relocatable object file, which section/part of the file contains Info for symbolic debugging, that is, a debugging symbol table with entries for local variables and typedefs defined in the program, global variables defined and referenced in the program, and the original C source file.

Select one:

- ☐ 1. .line section
- ☒ 2. .debug section ✓
- ☐ 3. .rel.text section
- ☐ 4. .rel.data section
- ☐ 5. Section header table
- ☐ 6. .symtab section
- ☐ 7. .strtab section

The correct answer is: .debug section

Given the following object codes for main and swap functions generated by the assembler which are going to be merged by the linker as shown below.

Please note that swap object code is aligned to quad or 8byte address boundary (to avoid segmentation fault due to address alignment violation) and placed after the main code.

Byte offset associated with the instructions assigned by the assembler is shown at the beginning of every instruction, followed by machine instruction and then assembly language instruction.

The main object code is located at byte offset 00000000 and swap object code is located at byte offset 00000018 hex.

00000000<main>:

Byte offset	Machine instruction	Assembly language
0:	55	push %ebp
1:	89 e5	mov %esp,%ebp
3:	83 ec 08	sub \$0x8,%esp
6:	e8 00 00 00 00	call swap
b:	31 c0	xor %eax,%eax
d:	89 ec	mov %ebp,%esp
f:	5d	pop %ebp
10:	c3	ret
11:	90	nop .align 8 (align swap to next quad address boundary)
12:	90	nop
13:	90	nop
14:	90	nop
15:	90	nop
16:	90	nop
17:	90	nop

00000018<swap>:

18:	55	push %ebp	
19:	8b 15 00 00 00 00	movl \$bufp0,%edx	Get *bufp0
1f:	c7 05 00 00 00 00	movl \$(bufp0+1), %eax	\$bufp1 = \$(bufp0+1)
25:	8b 0a	mov (%edx),%ecx	
27:	89 02	mov %eax,(%edx)	
29:	a1 00 00 00 00	mov \$bufp1, %eax	Get *bufp1
2e:	89 08	mov %ecx,(%eax)	
30:	5d	pop %ebp	
31:	c3	ret	

You are hereby required to provide the Relocation Entries for the references in the instructions at byte offset 19, clearly indicating the values of offset, relocation type, symbol and addend field values associated with each relocation entry.

Assume that all Call and jump instructions use PC-Relative addressing.

Select one:

- ☒ 1. 1b:R_X86_64_32 buf0 ✓
- ☐ 2. 1a:R_X86_64_32 buf0
- ☐ 3. 1b:R_X86_64_32 buf0 -0x4
- ☐ 4. 19:R_X86_64_32 buf0

The correct answer is: 1b:R_X86_64_32 buf0

Question **18**

Correct

Mark 10.00 out of 10.00

FPM stands for

Answer: Fast Page Mode



The correct answer is: Fast Page Mode

Question **19**

Correct

Mark 5.00 out of 5.00

..... reference patterns are a common and important source of spatial locality in programs.

Select one:

- ☐ 1. Stride-k
- ☒ 2. Stride-1 ✓
- ☐ 3. Stride-4
- ☐ 4. Stride-2

The correct answer is: Stride-1

Question **20**

Correct

Mark 5.00 out of 5.00

A Fully Associative Mapped Cache has

....., & set has

Select one:

- ☐ 1. MORE THAN ONE SET, ONE BLOCK/LINE
- ☒ 2. ONE SET, MORE THAN ONE BLOCK/LINE ✓
- ☐ 3. ONE SET, ONE BLOCK/LINE
- ☐ 4. MORE THAN ONE SET, MORE THAN ONE BLOCK/LINE

The correct answer is: ONE SET, MORE THAN ONE BLOCK/LINE

Question **21**

Correct

Mark 5.00 out of 5.00

On Linux tool is used to display the complete structure of an object file, including all of the information encoded in the ELF header.

Select one:

- ☒ 1. readelf ✓
- ☐ 2. nm
- ☐ 3. ldd
- ☐ 4. ar

The correct answer is: readelf

Question **22**

Correct

Mark 10.00 out of 10.00

Consider a 32-bit microprocessor that has an on-chip 64KB 4-way set associative cache.

Assume that the cache has a line (block) size of 4 words where each word is 32-bits (4 bytes).

Determine the number of bits in TAG, SET, BYTE OFFSET fields in the memory address.

Select one:

- ☒ 1. TAG=18 bits, INDEX=10 bits, BYTE OFFSET=4 bits ✓
- ☐ 2. TAG=20 bits, INDEX= 6 bits, BYTE OFFSET=6 bits
- ☐ 3. TAG=16 bits, INDEX=10 bits, BYTE OFFSET=6 bits
- ☐ 4. TAG=16 bits, INDEX=12 bits, BYTE OFFSET=4 bits

The correct answer is: TAG=18 bits, INDEX=10 bits, BYTE OFFSET=4 bits

Question 23

Not answered

Marked out of
20.00

The following program consists of two source files, main.c and swap.c. The main function initializes a two-element array of ints, and then calls the swap function to swap the pair.

```
/* main.c */
void swap();
int buf[2] = {1, 2};
int main() {
    swap();
    return 0;
}

/* swap.c */
extern int buf[];
int *bufp0 = &buf[0];
int *bufp1;
void swap() {
    int temp;
    bufp1 = &buf[1];
    temp = *bufp0;
    *bufp0 = *bufp1;
    *bufp1 = temp;
}
```

For each symbol that is defined or referenced in swap.o, indicate whether or not it will have a symbol table entry in the .symtab section in module swap.o.

If so, indicate the module (swap.o or main.o) that defines the symbol given below, the symbol type (local, global, or extern), and the section (.text, .data, or .bss) it occupies in that module.

Symbol : buf

Select one:

- ☐ 1. swap.o .symtab entry? :Yes, Symbol type : extern, Module where defined: main.o, Section: .data
- ☐ 2. swap.o .symtab entry? :Yes, Symbol type : global, Module where defined: swap.o, Section: .bss
- ☐ 3. swap.o .symtab entry? :Yes, Symbol type : global, Module where defined: swap.o, Section: .data
- ☐ 4. swap.o .symtab entry? :Yes, Symbol type : global, Module where defined: swap.o, Section: .text
- ☐ 5. swap.o .symtab entry? :no, Symbol type : -----, Module where defined: -----, Section: -----

The correct answer is: swap.o .symtab entry? :Yes, Symbol type : extern, Module where defined: main.o, Section: .data

Question **24**
Correct
Mark 10.00 out of 10.00

FULL FORM OF RAM

Answer: RANDOM ACCESS MEMORY



The correct answer is: RANDOM ACCESS MEMORY

Question **25**
Correct
Mark 5.00 out of 5.00

Global symbols that are referenced by module m but defined by some other module.

Select one:

- ☒ 1. External symbols ✓
- ☐ 2. Global symbols
- ☐ 3. Linker Local symbols
- ☐ 4. local program variables

The correct answer is: External symbols

Question **26**
Correct
Mark 10.00 out of 10.00

Identify the Kinds of Object Files

Select one:

- ☐ 1. Relocatable file (.o file)
- ☐ 2. Executable file (a.out file)
- ☐ 3. Shared file (.so file)
- ☒ 4. All of the above ✓

The correct answer is: All of the above

Question **27**
Correct
Mark 5.00 out of 5.00

..... is used to construct Cache memory

Select one:

- ☐ 1. EEPROM
- ☐ 2. DRAM
- ☐ 3. EPROM
- ☒ 4. SRAM ✓

The correct answer is: SRAM

Question **28**

Correct

Mark 10.00 out of 10.00

The sense circuit/ differential amplifier will produce a logic level on the data bus line, if there is difference between voltage levels on bit lines b & b' - Select more than one correct answer.

Select one or more:

- ☐ 1. 0, positive
- ☒ 2. 1, positive ✓
- ☒ 3. 0, negative ✓
- ☐ 4. 1, negative

The correct answers are: 1, positive, 0, negative

Question **29**

Incorrect

Mark 0.00 out of 10.00

Identify the following program file by clearly indicating which step of the Compiler driver has generated the file when the following gcc command is executed

`$gcc -o prog prog.c`

and also indicate type of the file (source file, intermediate C file, assembly language program, relocatable object file, Executable object file)

Program file : prog.s

Select one:

- ☒ 1. Generated by which compiler step :none, Input to which compiler tool:Preprocessor, Type of file :source file ✗
- ☐ 2. Generated by which compiler step :linker, Input to which compiler tool:none, Type of file :Executable object file
- ☐ 3. Generated by which compiler step :assembler, Input to which compiler tool:linker, Type of file :relocatable object file
- ☐ 4. Generated by which compiler step :Compiler, Input to which compiler tool:assembler, Type of file : assembly language program
- ☐ 5. Generated by which compiler step :Preprocessor, Input to which compiler tool:compiler, Type of file :intermediate C file

The correct answer is: Generated by which compiler step :Compiler, Input to which compiler tool:assembler, Type of file : assembly language program

Question 30

Not answered

Marked out of
20.00

The following program consists of two source files, main.c and swap.c. The main function initializes a two-element array of ints, and then calls the swap function to swap the pair.

```
/* main.c */
void swap();
int buf[2] = {1, 2};
int main() {
    swap();
    return 0;
}

/* swap.c */
extern int buf[];
int *bufp0 = &buf[0];
int *bufp1;
void swap() {
    int temp;
    bufp1 = &buf[1];
    temp = *bufp0;
    *bufp0 = *bufp1;
    *bufp1 = temp;
}
```

For each symbol that is defined or referenced in swap.o, indicate whether or not it will have a symbol table entry in the .symtab section in module swap.o.

If so, indicate the module that defines the symbol (swap.o or main.o), the symbol type (local, global, or extern), and the section (.text, .data, or .bss) it occupies in that module.

Symbol : temp

Select one:

- ☐ 1. swap.o .symtab entry? :no, Symbol type : -----, Module where defined: -----, Section:-----
- ☐ 2. swap.o .symtab entry? :Yes, Symbol type : global, Module where defined: swap.o, Section:.bss
- ☐ 3. swap.o .symtab entry? :Yes, Symbol type : extern, Module where defined: main.o, Section:.data
- ☐ 4. swap.o .symtab entry? :Yes, Symbol type : global, Module where defined: swap.o, Section:.data
- ☐ 5. swap.o .symtab entry? :Yes, Symbol type : global, Module where defined: swap.o, Section:.text

The correct answer is: swap.o .symtab entry? :no, Symbol type : -----, Module where defined: -----, Section:-----

Question **31**

Correct

Mark 5.00 out of 5.00

If the required word is not found in the cache then it is termed as

Select one:

- ☐ 1. Page Hit
- ☒ 2. Cache Miss ✓
- ☐ 3. Page Fault
- ☐ 4. Cache hit

The correct answer is: Cache Miss

Question **32**

Correct

Mark 10.00 out of 10.00

SSD stands for

Answer: Solid State Disks ✓

The correct answer is: Solid State Disks

Question **33**

Correct

Mark 5.00 out of 5.00

In DRAM, During a read or a Write operation, the row address is applied first and it is loaded into the row address latch, in response to a signal pulse on the input of the chip.

Select one:

- ☐ 1. Chip Select(CS)
- ☒ 2. Row Address Strobe (RAS) ✓
- ☐ 3. Read/Write (R/W)
- ☐ 4. Column Address Strobe (CAS)

The correct answer is: Row Address Strobe (RAS)

Question **34**

Correct

Mark 5.00 out of 5.00

On Linux, tool is used to create static libraries, and inserts, deletes, lists and extracts members

Select one:

- ☐ 1. ldd
- ☒ 2. ar ✓
- ☐ 3. readelf
- ☐ 4. nm

The correct answer is: ar

Question **35**

Correct

Mark 5.00 out of 5.00

A Direct Mapped Cache has

....., & each set has

Select one:

- ☐ 1. ONE SET, ONE BLOCK/LINE
- ☒ 2. MORE THAN ONE SET, ONE BLOCK/LINE ✓
- ☐ 3. MORE THAN ONE SET, MORE THAN ONE BLOCK/LINE
- ☐ 4. ONE SET, MORE THAN ONE BLOCK/LINE

The correct answer is: MORE THAN ONE SET, ONE BLOCK/LINE

Question **36**

Correct

Mark 5.00 out of 5.00

..... are Symbols defined by module m that can be referenced by other modules

Select one:

- ☒ 1. Global symbols ✓
- ☐ 2. local program variables
- ☐ 3. Linker Local symbols
- ☐ 4. External symbols

The correct answer is: Global symbols

Question **37**

Correct

Mark 5.00 out of 5.00

Number of bits or bytes that can be transferred in one second is referred to as

Answer:



The correct answer is: Memory Bandwidth

Question **38**

Correct

Mark 5.00 out of 5.00

Relocation entries for code are placed in by the assembler.

Select one:

- ☒ 1. .rel.text section ✓
- ☐ 2. .rodata section
- ☐ 3. .rel.data section
- ☐ 4. .bss section

The correct answer is: .rel.text section

Question **39**

Correct

Mark 10.00 out of 10.00

Whenever the assembler encounters a reference to an object whose ultimate location is unknown, it generates a that tells the linker how to modify the reference when it merges the object file into an executable.

Answer: relocation entry



The correct answer is: relocation entry

Question **40**

Correct

Mark 5.00 out of 5.00

Given the following program segment

```
int sumvec (int v[N])
```

```
{ int i, sum = 0;
```

```
  for (i = 0; i < n; i++)
```

```
    sum += v[i];
```

```
  return sum; }
```

This program exhibits with respect to both data and instruction accesses.

Select one:

- ☐ 1. only spatial locality
- ☒ 2. both spatial and temporal locality ✓
- ☐ 3. only temporal locality
- ☐ 4. no spatial and temporal locality

The correct answer is: both spatial and temporal locality

Given the following object codes for main and swap functions generated by the assembler which are going to be merged by the linker as shown below.

Please note that swap object code is aligned to quad or 8byte address boundary (to avoid segmentation fault due to address alignment violation) and placed after the main code.

Byte offset associated with the instructions assigned by the assembler is shown at the beginning of every instruction, followed by machine instruction and then assembly language instruction.

The main object code is located at byte offset 00000000 and swap object code is located at byte offset 00000018 hex.

00000000<main>:

Byte offset instruction	Machine instruction	Assembly language
0:	55	push %ebp
1:	89 e5	mov %esp,%ebp
3:	83 ec 08	sub \$0x8,%esp
6:	e8 00 00 00 00	call swap
b:	31 c0	xor %eax,%eax
d:	89 ec	mov %ebp,%esp
f:	5d	pop %ebp
10:	c3	ret
11:	90	nop .align 8 (align swap to next quad address boundary)
12:	90	nop
13:	90	nop
14:	90	nop
15:	90	nop
16:	90	nop
17:	90	nop

00000018<swap>:

18:	55	push %ebp	
19:	8b 15 00 00 00 00	movl \$bufp0,%edx	Get
*bufp0			
1f:	c7 05 00 00 00 00	movl \$(bufp0+1),	
%eax	\$bufp1 = \$(bufp0+1)		
25:	8b 0a	mov (%edx),%ecx	
27:	89 02	mov %eax,(%edx)	
29:	a1 00 00 00 00	mov \$bufp1, %eax	Get
*bufp1			
2e:	89 08	mov %ecx,(%eax)	
30:	5d	pop %ebp	
31:	c3	ret	

You are hereby required to provide the Relocation Entries for the references in the instructions at byte offset 29, clearly indicating the values of offset, relocation type, symbol and addend field values associated with each relocation entry.

Assume that all Call and jump instructions use PC-Relative addressing.

Select one:

- ☒ 1. 2a:R_X86_64_32 buf1 ✓
- ☐ 2. 29:R_X86_64_32 buf1 -0x4
- ☐ 3. 29:R_X86_64_32 buf1
- ☐ 4. 2a:R_X86_64_32 buf1 -0x4

The correct answer is: 2a:R_X86_64_32 buf1

Question **42**

Correct

Mark 10.00 out of 10.00

Function `execve` is

Select one:

- ☒ 1. called once, never returns ✓
- ☐ 2. called once, returns one or more times
- ☐ 3. called once, returns twice
- ☐ 4. called once, returns only once

The correct answer is: called once, never returns

Question **43**

Correct

Mark 5.00 out of 5.00

..... special pseudo section is used for symbols that should not be relocated.

Select one:

- ☐ 1. TEXT
- ☒ 2. ABS ✓
- ☐ 3. UND
- ☐ 4. COM

The correct answer is: ABS

Question **44**

Correct

Mark 5.00 out of 5.00

Identify the following Compiler Driver tool -Language preprocessor.

Select one:

- ☐ 1. as
- ☐ 2. ld
- ☒ 3. cpp ✓
- ☐ 4. cc1

The correct answer is: cpp

Question **45**

Correct

Mark 5.00 out of 5.00

Cache Memory is organized as a collection of one or more

Answer: ✓

The correct answer is: sets

Question **46**

Correct

Mark 5.00 out of 5.00

..... is a powerful linking technique that allows programmers to intercept calls to shared library functions and execute their own code instead.

Answer: Library interpositioning



The correct answer is: Library interpositioning

Question **47**

Correct

Mark 10.00 out of 10.00

DRAM STANDS FOR

Answer: DYNAMIC RANDOM ACCESS MEMORY



The correct answer is: DYNAMIC RAM

Question **48**

Correct

Mark 5.00 out of 5.00

Which one of the following cache mapping approach provides highest Hit ratio.

Select one:

- ☐ 1. 8-way Set Associative Cache
- ☒ 2. Fully Associative Cache
- ☐ 3. Direct Mapped Cache
- ☐ 4. 4-way Set associative Cache



The correct answer is: Fully Associative Cache

Question **49**

Correct

Mark 5.00 out of 5.00

Procedures and initialized globals are treated as symbols

Answer: Strong



The correct answer is: strong

Question **50**

Correct

Mark 5.00 out of 5.00

On Linux/Unix systems, static libraries are stored on disk in a particular file format known as an

Answer: archive



The correct answer is: archive

Question **51**
Correct
Mark 10.00 out of 10.00

PROM stands for

Answer: Programmable Read Only Memory ✓

The correct answer is: Programmable Read Only Memory

Question **52**
Correct
Mark 5.00 out of 5.00

In a program with good, if a memory location is referenced once, then the program is likely to reference a nearby memory location close together in time.

Answer: Spatial Locality ✓

The correct answer is: spatial locality

Question **53**
Correct
Mark 15.00 out of 15.00

Consider a 64-bit microprocessor that has an on-chip 128-KB 16-way Set-Associative mapped cache.

Assume that the cache has a line (block) size of 8 quad words where each quad word is 64-bits (8 bytes).

Determine the number of bits in TAG, SET, BYTE OFFSET fields in the memory address.

Select one:

- ☐ 1. TAG=52 bits, SET=8bits, BYTE OFFSET=4 bits
- ☐ 2. TAG=52 bits, SET=8bits, BYTE OFFSET=4 bits
- ☐ 3. TAG=50 bits, SET=10 bits, BYTE OFFSET=4 bits
- ☒ 4. TAG=51 bits, SET=7 bits, BYTE OFFSET=6 bits ✓

The correct answer is: TAG=51 bits, SET=7 bits, BYTE OFFSET=6 bits

Question **54**
Correct
Mark 5.00 out of 5.00

The purpose of step of the linker is to associate each symbol reference with exactly one symbol definition.

Answer: Symbol resolution ✓

The correct answer is: symbol resolution

Question **55**
Correct
Mark 10.00 out of 10.00

EDO stands for

Answer: Extended Data Out ✓

The correct answer is: Extended Data Out

Question **56**

Correct

Mark 5.00 out of 5.00

Static linker tool used on linux system is

Select one:

- ☐ 1. syscall
- ☐ 2. execve
- ☒ 3. ld ✓
- ☐ 4. ld-linux

The correct answer is: ld

Question **57**

Not answered

Marked out of 5.00

In a typical ELF(Executable and Linkable Format) relocatable object file, which section/part of the file contains a 16-byte sequence that describes the word size and byte ordering of the system that generated the file.

It also includes size of that part, the object file type (e.g., relocatable, executable, or shared), the machine type (e.g., IA32, x86-64), the file offset of the section header table, and the size and number of entries in the section header table.

Select one:

- ☐ 1. Segment header table
- ☐ 2. ELF header
- ☐ 3. .bss section
- ☐ 4. .rodata section
- ☐ 5. .text section
- ☐ 6. Section header table
- ☐ 7. .data section

The correct answer is: ELF header

Question **58**

Incorrect

Mark 0.00 out of 5.00

Given the following program segment

```
int sum_array_rows(int a[M][N])
{
    int i, j, sum = 0;

    for (j = 0; j < N; j++)
        for (i = 0; i < M; i++)
            sum += a[i][j];
    return sum;
}
```

This program exhibits with respect to both data and instruction accesses.

Select one:

- ☐ 1. temporal Locality, but no spatial locality
- ☐ 2. only spatial locality
- ☒ 3. both spatial and temporal locality ✖
- ☐ 4. no spatial and temporal locality

The correct answer is: temporal Locality, but no spatial locality

Question **59**

Not answered

Marked out of 10.00

Consider a 32-bit microprocessor that has an on-chip 64-KB 8-way Set-Associative mapped cache.

Assume that the cache has a line (block) size of 16 words where each word is 32-bits (4 bytes).

Determine the number of bits in TAG, SET, BYTE OFFSET fields in the memory address.

Select one:

- ☐ 1. TAG=19 bits, SET=7 bits, BYTE OFFSET=6 bits
- ☐ 2. TAG=19 bits, SET=9 bits, BYTE OFFSET=4 bits
- ☐ 3. TAG=20 bits, SET=8bits, BYTE OFFSET=4 bits
- ☐ 4. TAG=18 bits, SET=10 bits, BYTE OFFSET=4 bits

The correct answer is: TAG=19 bits, SET=7 bits, BYTE OFFSET=6 bits

Question **60**

Correct

Mark 5.00 out of 5.00

In policy, all writes are made only to the cache and write to the main memory is postponed until a replacement is needed.

Select one:

- ☐ 1. Write-through
- ☐ 2. Non-Allocate on write
- ☐ 3. Allocate on write
- ☒ 4. Write-back ✔

The correct answer is: Write-back

Question **61**

Correct

Mark 10.00 out of 10.00

Given a Computer System having a processor having clock cycle time 0.25ns with two levels of Caches and Main Memory with the following parameters:

L1 Cache : Hit rate 0.92, Hit Time 2 clock cycles

L2 Cache : Hit rate 0.98, Hit Time 4 clock cycles

Main Memory : Miss Penalty 75 clock cycles

Compute the Average Time for accessing the Cache block

Select one:

- ☒ 1. 0.61ns ✓
- ☐ 2. 0.71ns
- ☐ 3. 2.44ns
- ☐ 4. 0.51ns

The correct answer is: 0.61ns

Question **62**

Correct

Mark 10.00 out of 10.00

EEPROM stands for

Answer: ✓

The correct answer is: Electrically Erasable Programmable Read Only Memory

Question **63**

Correct

Mark 15.00 out of 15.00

Consider a 32-bit microprocessor that has an on-chip 64KB direct mapped cache.

Assume that the cache has a line (block) size of 4 words where each word is 32-bits (4 bytes).

Determine the number of bits in TAG, INDEX, BYTE OFFSET fields in the memory address.

Select one:

- ☐ 1. TAG=16 bits, INDEX=10 bits, BYTE OFFSET=6 bits
- ☐ 2. TAG=20 bits, INDEX= 6 bits, BYTE OFFSET=6 bits
- ☒ 3. TAG=16 bits, INDEX=12 bits, BYTE OFFSET=4 bits ✓
- ☐ 4. TAG=18 bits, INDEX=10 bits, BYTE OFFSET=4 bits

The correct answer is: TAG=16 bits, INDEX=12 bits, BYTE OFFSET=4 bits

Question 64

Correct

Mark 5.00 out of 5.00

Inpolicy, every write operation to the cache is repeated to the main memory at the same time

Select one:

- ☐ 1. Allocate on write
- ☐ 2. Write-back
- ☒ 3. Write-through ✓
- ☐ 4. Non-Allocate on write

The correct answer is: Write-through

Question 65

Correct

Mark 15.00 out of 15.00

Consider a 32-bit microprocessor that has an on-chip 64KB direct mapped cache.

Assume that the cache has a line (block) size of 4 words where each word is 32-bits (4 bytes).

Where in the cache is the word from memory location 79D5C789 mapped?

Select one:

- ☒ 1. TAG=0x79D5, INDEX=0xC78, BYTE OFFSET=0x09 ✓
- ☐ 2. TAG=0x79D5C, INDEX=0x01E, BYTE OFFSET=0x01
- ☐ 3. TAG=0x79D5, INDEX=0x078, BYTE OFFSET=0x09
- ☐ 4. TAG=0x79D5C, INDEX=0x078, BYTE OFFSET=0x09

The correct answer is: TAG=0x79D5, INDEX=0xC78, BYTE OFFSET=0x09

Question 66

Incorrect

Mark 0.00 out of 10.00

Identify the following program file by clearly indicating which step of the Compiler driver has generated the file when the following gcc command is executed

`$gcc -o prog prog.c`

and also indicate type of the file (source file, intermediate C file, assembly language program, relocatable object file, Executable object file)

Program file : prog.i

Select one:

- ☐ 1. Generated by which compiler step :Preprocessor, Input to which compiler tool:compiler, Type of file :intermediate C file
- ☐ 2. Generated by which compiler step :linker, Input to which compiler tool:none, Type of file :Executable object file
- ☒ 3. Generated by which compiler step :none, Input to which compiler tool:Preprocessor, Type of file :source file ✗
- ☐ 4. Generated by which compiler step :Compiler, Input to which compiler tool:assembler, Type of file : assembly language program
- ☐ 5. Generated by which compiler step :assembler, Input to which compiler tool:linker, Type of file :relocatable object file

The correct answer is: Generated by which compiler step :Preprocessor, Input to which compiler tool:compiler, Type of file :intermediate C file

Question 67

Correct

Mark 50.00 out of 50.00

Given the following version of the Matrix Multiplication program.

Assume that matrices a, b & c containing elements of the type double are stored in the main memory in row-major order.

The size of the cache block is 8 elements where each element is a double, i.e., size of each element is 8 bytes

Assume that memory system has one level of cache using write-through policy and r, i, j, k are all mapped to registers

```
/* jki */
for (j=0; j<80; j++) {
    for (k=0; k<80; k++) {
        r = b[k][j];
        for (i=0; i<80; i++)
            c[i][j] += a[i][k] * r;
    }
}
```

Determine the total references T in the innermost loop i considering all the i elements accessed where $i = 0, \dots, 79$.

Determine the total number of misses M in the innermost loop i considering all the i elements accessed where $i = 0, \dots, 79$.

Determine the total number of Hits H in the innermost loop i considering all the i elements accessed where $i = 0, \dots, 79$.

Determine the Hit Rate h for the total references in the innermost loop i

Determine the Miss Rate (1-h) for the total references in the innermost loop i

Determine the actual total number of loads L considering each memory read miss as one load

Determine the actual total number of stores S considering each memory write as one store

Select one:

- ☐ 1. T=80, M=80, H=0, h=0, (1-h)=15, L=80, S=0
- ☒ 2. T=160, M=160, H=0, h=0, (1-h)=1, L=160, S=80 ✓
- ☐ 3. T=160, M=160, H=0, h=0, (1-h)=1, L=160, S=160
- ☐ 4. T=80, M=0, H=80, h=1, (1-h)=0, L=0, S=80

The correct answer is: T=160, M=160, H=0, h=0, (1-h)=1, L=160, S=80

Question **68**

Incorrect

Mark 0.00 out of 15.00

Given a Seagate Cheetah 15K.4 Hard disk drive containing 4 platters having a zone map in the table shown below.

Determine the number of spare cylinders in the Zone 10.

Zone Number	Sectors per track	Cylinders per zone	Logical Blocks per zone
0	864	3201	22,076,928
1	844	3200	21,559,136
2	816	3400	22,149,504
3	806	3100	19,943,664
4	795	3100	19,671,480
5	768	3400	20,852,736
6	760	3450	21,159,936
7	725	3650	21,135,200
8	704	3700	20,804,608
9	672	3700	19,858,944
10	640	3700	18,913,280
11	603	3700	17,819,856
12	576	3707	17,054,208
13	528	3060	12,900,096

Select one:

- ☒ 1. 7 
- ☐ 2. 8
- ☐ 3. 6
- ☐ 4. 9

The correct answer is: 6

Question **69**

Correct

Mark 5.00 out of 5.00

.....refers to the amount of time it takes to transfer a word of data to or from the memory.

Answer:



The correct answer is: Memory latency

Question **70**

Correct

Mark 5.00 out of 5.00

In a typical ELF(Executable and Linkable Format) relocatable object file, which section/part of the file contains a string table for the symbol tables in the .symtab and .debug sections and for section names in the section headers

Select one:

- ☐ 1. .rel.text section
- ☐ 2. .symtab section
- ☐ 3. Section header table
- ☐ 4. .rel.data section
- ☒ 5. .strtab section ✓
- ☐ 6. .debug section
- ☐ 7. .line section

The correct answer is: .strtab section

Question **71**

Correct

Mark 10.00 out of 10.00

Identify the following program file by clearly indicating which step of the Compiler driver has generated the file when the following gcc command is executed

`$gcc -o prog prog.c`

and also indicate type of the file(source file, intermediate C file, assembly language program, relocatable object file, Executable object file

Program file : prog.c

Select one:

- ☐ 1. Generated by which compiler step :linker, Input to which compiler tool:none, Type of file :Executable object file
- ☐ 2. Generated by which compiler step :Preprocessor, Input to which compiler tool:compiler, Type of file :intermediate C file
- ☒ 3. Generated by which compiler step :none, Input to which compiler tool:Preprocessor, Type of file :source file ✓
- ☐ 4. Generated by which compiler step :assembler, Input to which compiler tool:linker, Type of file :relocatable object file
- ☐ 5. Generated by which compiler step :Compiler, Input to which compiler tool:assembler, Type of file : assembly language program

The correct answer is: Generated by which compiler step :none, Input to which compiler tool:Preprocessor, Type of file :source file

Question **72**

Correct

Mark 5.00 out of 5.00

Indicate the correct order of the steps in the Compilation process

Select one:

- ☐ 1. compiler, language preprocessor, linker, and assembler
- ☐ 2. compiler, language preprocessor, assembler, and linker
- ☐ 3. language preprocessor, assembler, compiler, and linker
- ☒ 4. language preprocessor, compiler, assembler, and linker ✓

The correct answer is: language preprocessor, compiler, assembler, and linker

Question **73**

Correct

Mark 5.00 out of 5.00

.....,, Occurs when the set of active cache blocks is larger than the cache(i.e., cache is full).

Select one:

- ☐ 1. Cold(Compulsory) miss
- ☐ 2. Conflict miss
- ☒ 3. Capacity miss ✓
- ☐ 4. TBL miss

The correct answer is: Capacity miss

Question **74**

Incorrect

Mark 0.00 out of 5.00

Provide the name of the executable object file generated by the following command :

`gcc -o sum sum.c`

Answer:



The correct answer is: sum

Question **75**

Correct

Mark 10.00 out of 10.00

Identify the Linker symbols in the following :

Select one:

- ☐ 1. Global symbols
- ☐ 2. External symbols
- ☐ 3. Linker Local symbols
- ☒ 4. All of the above ✓

The correct answer is: All of the above

Question **76**

Correct

Mark 5.00 out of 5.00

Linking performed at compile time, when the source code is translated into machine code is termed as

Answer: ✓

The correct answer is: STATIC LINKING

Question **77**

Correct

Mark 5.00 out of 5.00

In Linux, dynamic linker interface function takes a handle to a previously opened shared library and a symbol name, and returns the address of the symbol, if it exists, or NULL otherwise.

Select one:

- ☐ 1. dlclose
- ☒ 2. dlsym ✓
- ☐ 3. dlerror
- ☐ 4. dlopen

The correct answer is: dlsym

Question **78**

Correct

Mark 10.00 out of 10.00

DIMM stands for

Answer: ✓

The correct answer is: Dual Inline Memory Module

Question **79**

Correct

Mark 5.00 out of 5.00

In a typical ELF(Executable and Linkable Format) relocatable object file, which section/part of the file contains Relocation info for .text section, that is,

A list of locations (symbol references) in the .text section that will need to be modified when the linker combines this object file with others.

Select one:

- ☐ 1. .line section
- ☐ 2. Section header table
- ☐ 3. .debug section
- ☒ 4. .rel.text section ✓
- ☐ 5. .symtab section
- ☐ 6. .rel.data section
- ☐ 7. .strtab section

The correct answer is: .rel.text section

Given the following object codes for main and swap functions generated by the assembler which are going to be merged by the linker as shown below.

Please note that swap object code is aligned to quad or 8byte address boundary (to avoid segmentation fault due to address alignment violation) and placed after the main code.

Byte offset associated with the instructions assigned by the assembler is shown at the beginning of every instruction, followed by machine instruction and then assembly language instruction.

The main object code is located at byte offset 00000000 and swap object code is located at byte offset 00000018 hex.

00000000<main>:

Byte offset instruction	Machine instruction	Assembly language
0:	55	push %ebp
1:	89 e5	mov %esp,%ebp
3:	83 ec 08	sub \$0x8,%esp
6:	e8 00 00 00 00	call swap
b:	31 c0	xor %eax,%eax
d:	89 ec	mov %ebp,%esp
f:	5d	pop %ebp
10:	c3	ret
11:	90	nop .align 8 (align swap to next quad address boundary)
12:	90	nop
13:	90	nop
14:	90	nop
15:	90	nop
16:	90	nop
17:	90	nop

00000018<swap>:

18:	55	push %ebp	
19:	8b 15 00 00 00 00	movl \$bufp0,%edx	Get
*bufp0			
1f:	c7 05 00 00 00 00	movl \$(bufp0+1),	
%eax	\$bufp1 = \$(bufp0+1)		
25:	8b 0a	mov (%edx),%ecx	
27:	89 02	mov %eax,(%edx)	
29:	a1 00 00 00 00	mov \$bufp1, %eax	Get
*bufp1			
2e:	89 08	mov %ecx,(%eax)	
30:	5d	pop %ebp	
31:	c3	ret	

You are hereby required to provide the Relocation Entries for the references in the instructions at byte offset 6, clearly indicating the values of offset, relocation type, symbol and addend field values associated with each relocation entry.

Assume that all Call and jump instructions use PC-Relative addressing.

Select one:

- ☐ 1. 06:R_X86_64_PC32 swap -0x5
- ☐ 2. 07:R_X86_64_PC32 swap 0x4
- ☒ 3. 07:R_X86_64_PC32 swap -0x4 ✓
- ☐ 4. 06:R_X86_64_PC32 swap 0x5

The correct answer is: 07:R_X86_64_PC32 swap -0x4

Question **81**

Correct

Mark 5.00 out of 5.00

On Linux, tool is used to list the shared libraries that an executable needs at run time.

Select one:

- ☐ 1. nm
- ☒ 2. ldd ✓
- ☐ 3. readelf
- ☐ 4. ar

The correct answer is: ldd

Question **82**

Correct

Mark 5.00 out of 5.00

In Cache memory, each Set is a collection of one or more

Answer: blocks or lines ✓

The correct answer is: blocks or lines

Question **83**

Correct

Mark 5.00 out of 5.00

Linking performed at load time, when the program is loaded into memory is termed as

Answer: DYNAMIC LINKING ✓

The correct answer is: DYNAMIC LINKING

Question **84**

Correct

Mark 5.00 out of 5.00

..... special pseudo section is used for undefined symbols, that is, symbols that are referenced in this object module but defined elsewhere.

Select one:

- ☐ 1. COM
- ☒ 2. UND ✓
- ☐ 3. ABS
- ☐ 4. TEXT

The correct answer is: UND

Question **85**

Correct

Mark 5.00 out of 5.00

..... function is used on the Linux/Unix system to invoke the loader to load the executable into memory and execute.

Select one:

- ☒ 1. execve ✓
- ☐ 2. ld-linux
- ☐ 3. ld
- ☐ 4. syscall

The correct answer is: execve

Question **86**

Correct

Mark 10.00 out of 10.00

FULL FORM OF ROM

Answer: ✓

The correct answer is: READ ONLY MEMORY

Question **87**

Correct

Mark 10.00 out of 10.00

Library Interpositioning can occur at

Select one:

- ☐ 1. Compile time
- ☐ 2. Link time
- ☐ 3. Load/run time
- ☒ 4. All of the above ✓

The correct answer is: All of the above

Question **88**

Correct

Mark 5.00 out of 5.00

..... special pseudo section is used for uninitialized data objects that are not yet allocated.

Select one:

- ☒ 1. COM ✓
- ☐ 2. TEXT
- ☐ 3. UND
- ☐ 4. ABS

The correct answer is: COM

Question **89**

Incorrect

Mark 0.00 out
of 10.00

Identify the following program file by clearly indicating which step of the Compiler driver has generated the file when the following gcc command is executed

```
$gcc -o prog prog.c
```

and also indicate type of the file (source file, intermediate C file, assembly language program, relocatable object file, Executable object file)

Program file : prog.o

Select one:

- ☐ 1. Generated by which compiler step :Preprocessor, Input to which compiler tool:compiler, Type of file :intermediate C file
- ☐ 2. Generated by which compiler step :Compiler, Input to which compiler tool:assembler, Type of file : assembly language program
- ☐ 3. Generated by which compiler step :assembler, Input to which compiler tool:linker, Type of file :relocatable object file
- ☐ 4. Generated by which compiler step :linker, Input to which compiler tool:none, Type of file :Executable object file
- ☒ 5. Generated by which compiler step :none, Input to which compiler tool:Preprocessor, Type of file :source file ❌

The correct answer is: Generated by which compiler step :assembler, Input to which compiler tool:linker, Type of file :relocatable object file

Question 90

Correct

Mark 10.00 out of 10.00

Can you permute the loops so that the function scans the 3-d array *a* with a stride-1 reference pattern (and thus has good spatial locality)?

```
int sum_array_3d(int a[M][N][N])
{
    int i, j, k, sum = 0;

    for (i = 0; i < M; i++)
        for (j = 0; j < N; j++)
            for (k = 0; k < N; k++)
                sum += a[k][i][j];
    return sum;
}
```

Select one:

- ☐ 1.

```
int sum_array_3d(int a[M][N][N])
{
    int i, j, k, sum = 0;

    for (j = 0; j < N; j++)
        for (i = 0; i < M; i++)
            for (k = 0; k < N; k++)
                sum += a[k][i][j];
    return sum;
}
```
- ☐ 2.

```
int sum_array_3d(int a[M][N][N])
{
    int i, j, k, sum = 0;

    for (j = 0; j < N; j++)
        for (k = 0; k < N; k++)
            for (i = 0; i < M; i++)
                sum += a[k][i][j];
    return sum;
}
```
- ☒ 3.

```
int sum_array_3d(int a[M][N][N])
{
    int i, j, k, sum = 0;

    for (i = 0; i < M; i++)
        for (j = 0; j < N; j++)
            for (k = 0; k < N; k++)
                sum += a[i][j][k];
    return sum;
}
```

 ✓
- ☐ 4.

```
int sum_array_3d(int a[M][N][N])
{
    int i, j, k, sum = 0;

    for (k = 0; k < N; k++)
        for (j = 0; j < N; j++)
            for (i = 0; i < M; i++)
                sum += a[k][i][j];
    return sum;
}
```

The correct answer is:

```
int sum_array_3d(int a[M][N][N])
```

```
{
    int i, j, k, sum = 0;
```

```

for (i = 0; i < M; i++)
    for (j = 0; j < N; j++)
        for (k = 0; k < N; k++)
            sum += a[i][j][k];
return sum;
}

```

Question **91**

Correct

Mark 20.00 out of 20.00

Given the following statements in an relocatable object file generated by an assembler:

```

9: bf 00 00 00 00    mov  $0x0,%edi    # %edi = &array
a: R_X86_64_32 array      # Relocation entry

```

Determine the following fields in the Relocation entry : r.offset, r.type, r.symbol, r.addend

Select one:

- ☐ 1. r.offset = 0, r.type = R_X86_64_32, r.symbol = array, r.addend = 0xa
- ☐ 2. r.offset = -0x4, r.type = R_X86_64_32, r.symbol = array, r.addend = -0xa
- ☐ 3. r.offset = 0xa, r.type = R_X86_64_32, r.symbol = array, r.addend = -0x4
- ☒ 4. r.offset = 0xa, r.type = R_X86_64_32, r.symbol = array, r.addend = 0 ✓

The correct answer is: r.offset = 0xa, r.type = R_X86_64_32, r.symbol = array, r.addend = 0

Given the following object codes for main and swap functions generated by the assembler which are going to be merged by the linker as shown below.

Please note that swap object code is aligned to quad or 8byte address boundary (to avoid segmentation fault due to address alignment violation) and placed after the main code.

Byte offset associated with the instructions assigned by the assembler is shown at the beginning of every instruction, followed by machine instruction and then assembly language instruction.

The main object code is located at byte offset 00000000 and swap object code is located at byte offset 00000018 hex.

00000000<main>:

Byte offset instruction	Machine instruction	Assembly language
0:	55	push %ebp
1:	89 e5	mov %esp,%ebp
3:	83 ec 08	sub \$0x8,%esp
6:	e8 00 00 00 00	call swap
b:	31 c0	xor %eax,%eax
d:	89 ec	mov %ebp,%esp
f:	5d	pop %ebp
10:	c3	ret
11:	90	nop .align 8 (align swap to next quad address boundary)
12:	90	nop
13:	90	nop
14:	90	nop
15:	90	nop
16:	90	nop
17:	90	nop

00000018<swap>:

18:	55	push %ebp	
19:	8b 15 00 00 00 00	movl \$bufp0,%edx	Get
*bufp0			
1f:	c7 05 00 00 00 00	movl \$(bufp0+1),	
%eax	\$bufp1 = \$(bufp0+1)		
25:	8b 0a	mov (%edx),%ecx	
27:	89 02	mov %eax,(%edx)	
29:	a1 00 00 00 00	mov \$bufp1, %eax	Get
*bufp1			
2e:	89 08	mov %ecx,(%eax)	
30:	5d	pop %ebp	
31:	c3	ret	

You are hereby required to provide the Relocation Entries for the references in the instructions at byte offset 1f, clearly indicating the values of offset, relocation type, symbol and addend field values associated with each relocation entry.

Assume that all Call and jump instructions use PC-Relative addressing.

Select one:

- ☐ 1. 21:R_X86_64_32 buf0
- ☐ 2. 1f:R_X86_64_32 buf0 +0x4
- ☐ 3. 20:R_X86_64_32 buf0 +0x4
- ☒ 4. 21:R_X86_64_32 buf0 +0x4 ✓

The correct answer is: 21:R_X86_64_32 buf0 +0x4

Question **93**

Correct

Mark 10.00 out
of 10.00

List all the strong and weak symbols in the following code :

```
/* swap.c */  
extern int buf[];  
int *bufp0 = &buf[0];  
int *bufp1;  
void swap() {  
    int temp;  
    bufp1 = &buf[1];  
    temp = *bufp0;  
    *bufp0 = *bufp1;  
    *bufp1 = temp;  
}
```

Select one:

- ☒ 1. Strong Symbols :bufp0, swap, Weak Symbols: buf, bufp1 ✓
- ☐ 2. Strong Symbols :bufp0, swap, temp, Weak Symbols: buf, bufp1
- ☐ 3. Strong Symbols :bufp1, swap, Weak Symbols: buf, bufp0, temp
- ☐ 4. Strong Symbols :bufp0, swap, Weak Symbols: buf, bufp1,temp

The correct answer is: Strong Symbols :bufp0, swap, Weak Symbols: buf, bufp1

Question 94

Correct

Mark 15.00 out of 15.00

Suppose that a 16 MB file consisting of 512-byte logical blocks is stored on a disk drive with the following characteristics:

Parameter	Value
-----	-----
Rotational rate	10,000 RPM
Tavg seek	5 ms
Average # sectors/track	1024
Platters	16
Sector size	512 bytes

For the best case below, suppose that a program reads the logical blocks of the file sequentially, one after the other, and that the time to position the head over the first block is Tavg seek + Tavg rotation.

Best case: Estimate the optimal time (in ms) required to read the file given the best possible mapping of logical blocks to disk sectors (i.e., sequential).

Select one:

- ☒ 1. 200ms ✓
- ☐ 2. 192ms
- ☐ 3. 96ms
- ☐ 4. 100ms

The correct answer is: 200ms

Question 95

Not answered

Marked out of 5.00

In a typical ELF(Executable and Linkable Format) relocatable object file, which section/part of the file contains relocation entries required to resolve symbol references in Code section.

Select one:

- ☐ 1. .symtab section
- ☐ 2. .line section
- ☐ 3. .debug section
- ☐ 4. .strtab section
- ☐ 5. Section header table
- ☐ 6. .rel.text section
- ☐ 7. .rel.data section

The correct answer is: .rel.text section

Question **96**
Correct
Mark 10.00 out of 10.00

Given a Computer System having a processor with two levels of Caches and Main Memory with the following parameters:

L1 Cache : Hit Rate h_{L1} , Hit Time ht_{L1} (in clock cycles)

L2 Cache : Hit Rate h_{L2} , Hit Time ht_{L2} (in clock cycles)

Main Memory : Miss Penalty mp (in clock cycles)

Write the expression for computing the Average Time for accessing the Cache block

Select one:

- ☐ 1. Average Access Time $Tav = h_{L1} + (1 - h_{L1})(h_{L2} + (1 - h_{L2}) * mp)$
- ☐ 2. Average Access Time $Tav = ht_{L1} * (1 - h_{L1})(ht_{L2} + (1 - h_{L2}) * mp)$
- ☒ 3. Average Access Time $Tav = ht_{L1} + (1 - h_{L1})(ht_{L2} + (1 - h_{L2}) * mp)$ ✓
- ☐ 4. Average Access Time $Tav = ht_{L1} + (1 - ht_{L1})(ht_{L2} + (1 - ht_{L2}) * mp)$

The correct answer is: Average Access Time $Tav = ht_{L1} + (1 - h_{L1})(ht_{L2} + (1 - h_{L2}) * mp)$

Question **97**
Correct
Mark 5.00 out of 5.00

..... write hit policy uses Allocate on write upon a cache miss.

Select one:

- ☐ 1. Write-through
- ☐ 2. Non-Allocate on write
- ☐ 3. Allocate on write
- ☒ 4. Write-back ✓

The correct answer is: Write-back

Question **98**
Correct
Mark 5.00 out of 5.00

Modern Unix systems—such as Linux, later versions of System V Unix, BSD Unix variants, and SunSolaris use the format to store the object files.

Select one:

- ☐ 1. Common Object File format (COFF)
- ☐ 2. a.out
- ☐ 3. Portable Executable (PE)
- ☒ 4. Executable and Linkable Format (ELF) ✓

The correct answer is: Executable and Linkable Format (ELF)

Question 99
Correct
Mark 20.00 out
of 20.00

The following program consists of two source files, main.c and swap.c. The main function initializes a two-element array of ints, and then calls the swap function to swap the pair.

```
/* main.c */  
void swap();  
int buf[2] = {1, 2};  
int main() {  
    swap();  
    return 0;  
}  
  
/* swap.c */  
extern int buf[];  
int *bufp0 = &buf[0];  
int *bufp1;  
void swap() {  
    int temp;  
    bufp1 = &buf[1];  
    temp = *bufp0;  
    *bufp0 = *bufp1;  
    *bufp1 = temp;  
}
```

For each symbol that is defined or referenced in swap.o, indicate whether or not it will have a symbol table entry in the .symtab section in module swap.o.

If so, indicate the module (swap.o or main.o) that defines the symbol given below, the symbol type (local, global, or extern), and the section (.text, .data, or .bss) it occupies in that module.

Symbol : swap

Select one:

- ☒ 1. swap.o .symtab entry? :Yes, Symbol type : global, Module where defined: swap.o, Section: .text ✓
- ☐ 2. swap.o .symtab entry? :Yes, Symbol type : extern, Module where defined: main.o, Section: .data
- ☐ 3. swap.o .symtab entry? :Yes, Symbol type : global, Module where defined: swap.o, Section: .bss
- ☐ 4. swap.o .symtab entry? :Yes, Symbol type : global, Module where defined: swap.o, Section: .data
- ☐ 5. swap.o .symtab entry? :no, Symbol type : -----, Module where defined: -----, Section: -----

The correct answer is: swap.o .symtab entry? :Yes, Symbol type : global, Module where defined: swap.o, Section: .text

Question **100**

Incorrect

Mark 0.00 out of 10.00

EPROM stands for

Answer: ❌

The correct answer is: Erasable Programmable Read Only Memory

Question **101**

Correct

Mark 5.00 out of 5.00

In a typical ELF(Executable and Linkable Format) relocatable object file, which section/part of the file contains information such as Page size, virtual addresses memory segments (sections), segment sizes.

Select one:

- ☐ 1. .text section
- ☐ 2. .data section
- ☐ 3. ELF header
- ☐ 4. .rodata section
- ☒ 5. Segment header table ✔

The correct answer is: Segment header table

Question **102**

Incorrect

Mark 0.00 out of 10.00

Identify the following program file by clearly indicating which step of the Compiler driver has generated the file when the following gcc command is executed

\$gcc -o prog prog.c

and also indicate type of the file(source file, intermediate C file, assembly language program, relocatable object file, Executable object file

Program file : prog

Select one:

- ☐ 1. Generated by which compiler step :Compiler, Input to which compiler tool:assembler, Type of file : assembly language program
- ☐ 2. Generated by which compiler step :none, Input to which compiler tool:Preprocessor, Type of file :source file
- ☐ 3. Generated by which compiler step :linker, Input to which compiler tool:none, Type of file :Executable object file
- ☐ 4. Generated by which compiler step :Preprocessor, Input to which compiler tool:compiler, Type of file :intermediate C file
- ☒ 5. Generated by which compiler step :assembler, Input to which compiler tool:linker, Type of file :relocatable object file ❌

The correct answer is: Generated by which compiler step :linker, Input to which compiler tool:none, Type of file :Executable object file

Question **103**

Incorrect

Mark 0.00 out of 5.00

In amemory device, data is retained in the memory as long as power is applied, once power is removed the data stored gets lost.

Answer: 2,4,1,6,3,5



The correct answer is: VOLATILE

Question **104**

Correct

Mark 5.00 out of 5.00

In a typical ELF(Executable and Linkable Format) relocatable object file, which section/part of the file contains Read only data such as format strings in printf statements & jump tables for switch statements.

Select one:

- ☐ 1. Segment header table
- ☐ 2. Section header table
- ☐ 3. .data section
- ☐ 4. ELF header
- ☒ 5. .rodata section ✓
- ☐ 6. .bss section
- ☐ 7. .text section

The correct answer is: .rodata section

Question **105**

Correct

Mark 5.00 out of 5.00

In a typical ELF(Executable and Linkable Format) relocatable object file, which section/part of the file contains Relocation info for .data section, Addresses of pointer data that will need to be modified in the merged executable.

Select one:

- ☐ 1. .strtab section
- ☐ 2. Section header table
- ☐ 3. .line section
- ☐ 4. .symtab section
- ☐ 5. .debug section
- ☐ 6. .rel.text section
- ☒ 7. .rel.data section ✓

The correct answer is: .rel.data section

Question **106**

Correct

Mark 10.00 out of 10.00

local C variables are stored on the stack.

Answer: non-static



The correct answer is: non-static

Question **107**

Correct

Mark 10.00 out of 10.00

VRAM stands for

Answer: Video RAM



The correct answer is: Video RAM

Question **108**

Correct

Mark 5.00 out of 5.00

....., occurs when the cache is large enough, but multiple data objects all map to the same block in the cache.

Select one:

- ☐ 1. Capacity miss
- ☒ 2. Conflict miss
- ☐ 3. TBL miss
- ☐ 4. Cold(Compulsory) miss

The correct answer is: Conflict miss

Question **109**

Correct

Mark 10.00 out of 10.00

DDR STANDS FOR

Answer: Double Data Rate



The correct answer is: DOUBLE DATA RATE

Question **110**

Correct

Mark 5.00 out of 5.00

Linking performed at run time is termed as

Answer: Dynamic Linking



The correct answer is: DYNAMIC LINKING

Question **111**

Correct

Mark 5.00 out of 5.00

Symbols that are defined and referenced exclusively by module m.

Select one:

- ☐ 1. local program variables
- ☒ 2. Linker Local symbols
- ☐ 3. Global symbols
- ☐ 4. External symbols

The correct answer is: Linker Local symbols

Question **112**

Correct

Mark 5.00 out of 5.00

In Write-back policy, every cache block is assigned a bit, called the, to indicate that atleast one write operation has been made to the block while residing in the cache.

Select one:

- ☐ 1. valid bit
- ☐ 2. execute bit
- ☒ 3. dirty bit ✓
- ☐ 4. protection bit

The correct answer is: dirty bit

Question **113**

Correct

Mark 5.00 out of 5.00

Uninitialized globals are treated as symbols

Answer: ✓

The correct answer is: weak

Question **114**

Correct

Mark 5.00 out of 5.00

Examples of Linker Local symbols are

Select one:

- ☐ 1. non-static C functions and non-static global variables
- ☐ 2. C functions and global variables declared with the extern attribute
- ☒ 3. C functions and global variables defined with the static attribute ✓
- ☐ 4. local program variables

The correct answer is: C functions and global variables defined with the static attribute

Question **115**

Correct

Mark 5.00 out of 5.00

In Linux, dynamic linker interface function loads and links the shared library filename.

Select one:

- ☒ 1. dlopen ✓
- ☐ 2. dlerror
- ☐ 3. dlclose
- ☐ 4. dlsym

The correct answer is: dlopen

Question **116**

Correct

Mark 15.00 out of 15.00

Consider a 32-bit microprocessor that has an on-chip 64-KB 8-way Set-Associative mapped cache.

Assume that the cache has a line (block) size of 16 words where each word is 32-bits (4 bytes).

Where in the cache is the word from memory location FEEDBAC8 mapped?

Select one:

- ☐ 1. TAG=0x7F76D, SET=0x1AC, BYTE OFFSET=0x08
- ☐ 2. TAG=0xFEEDB, SET=0xAC, BYTE OFFSET=0x08
- ☒ 3. TAG=0x7F76D, SET=0x6B, BYTE OFFSET=0x08 ✓
- ☐ 4. TAG=0x3FBB6, SET=0x3AC, BYTE OFFSET=0x08

The correct answer is: TAG=0x7F76D, SET=0x6B, BYTE OFFSET=0x08

Question 117

Correct

Mark 20.00 out of 20.00

The following program consists of two source files, main.c and swap.c. The main function initializes a two-element array of ints, and then calls the swap function to swap the pair.

```
/* main.c */
void swap();
int buf[2] = {1, 2};
int main() {
    swap();
    return 0;
}

/* swap.c */
extern int buf[];
int *bufp0 = &buf[0];
int *bufp1;
void swap() {
    int temp;
    bufp1 = &buf[1];
    temp = *bufp0;
    *bufp0 = *bufp1;
    *bufp1 = temp;
}
```

For each symbol that is defined or referenced in swap.o, indicate whether or not it will have a symbol table entry in the .symtab section in module swap.o.

If so, indicate the module (swap.o or main.o) that defines the symbol given below, the symbol type (local, global, or extern), and the section (.text, .data, or .bss) it occupies in that module.

Symbol : bufp0

Select one:

- ☐ 1. swap.o .symtab entry? :Yes, Symbol type : global, Module where defined: swap.o, Section: .bss
- ☒ 2. swap.o .symtab entry? :Yes, Symbol type : global, Module where defined: swap.o, Section: .data ✓
- ☐ 3. swap.o .symtab entry? :Yes, Symbol type : global, Module where defined: swap.o, Section: .text
- ☐ 4. swap.o .symtab entry? :no, Symbol type : -----, Module where defined: -----, Section:-----
- ☐ 5. swap.o .symtab entry? :Yes, Symbol type : extern, Module where defined: main.o, Section: .data

The correct answer is: swap.o .symtab entry? :Yes, Symbol type : global, Module where defined: swap.o, Section: .data

Question **118**

Correct

Mark 5.00 out of 5.00

Standard DIMM have data path

Select one:

- ☒ 1. 64-bit ✓
- ☐ 2. 16-bit
- ☐ 3. 8-bit
- ☐ 4. 32-bit

The correct answer is: 64-bit

Question **119**

Correct

Mark 5.00 out of 5.00

Examples of Linker External symbols are

Select one:

- ☐ 1. C functions and global variables defined with the static attribute
- ☒ 2. C functions and global variables declared with the extern attribute ✓
- ☐ 3. local program variables
- ☐ 4. non-static C functions and non-static global variables

The correct answer is: C functions and global variables declared with the extern attribute

Question **120**

Correct

Mark 10.00 out of 10.00

Firmware on the controller performs a fast table lookup that translates the into a (surface, track, sector) triple that uniquely identifies the corresponding physical sector.

Answer: ✓

The correct answer is: logical block number

Question 121

Incorrect

Mark 0.00 out of 15.00

Given a Seagate Cheetah 15K.4 Hard disk drive containing 4 platters having a zone map in the table shown below.

Determine the number of spare cylinders in the Zone 0.

Zone Number	Sectors per track	Cylinders per zone	Logical Blocks per zone
0	864	3201	22,076,928
1	844	3200	21,559,136
2	816	3400	22,149,504
3	806	3100	19,943,664
4	795	3100	19,671,480
5	768	3400	20,852,736
6	760	3450	21,159,936
7	725	3650	21,135,200
8	704	3700	20,804,608
9	672	3700	19,858,944
10	640	3700	18,913,280
11	603	3700	17,819,856
12	576	3707	17,054,208
13	528	3060	12,900,096

Select one:

- ☒ 1. 6 ✖
- ☐ 2. 8
- ☐ 3. 7
- ☐ 4. 9

The correct answer is: 7

Question 122

Correct

Mark 5.00 out of 5.00

In Write-back Policy, at replacement time, the is checked, if it is set, then the block is written back to the main memory, otherwise it is simply overwritten by the incoming block.

Select one:

- ☐ 1. valid bit
- ☐ 2. execute bit
- ☐ 3. protection bit
- ☒ 4. dirty bit ✔

The correct answer is: dirty bit

Question 123

Correct

Mark 5.00 out of 5.00

Given the following program segment with respect to both data and instruction accesses.

```
int sum_array_rows(int a[M][N])
{
    int i, j, sum = 0;

    for (i = 0; i < M; i++)
        for (j = 0; j < N; j++)
            sum += a[i][j];
    return sum;
}
```

This program exhibits with respect to both data and instruction accesses.

Select one:

- ☐ 1. no spatial and temporal locality
- ☐ 2. only spatial locality
- ☒ 3. both spatial and temporal locality ✓
- ☐ 4. only temporal locality

The correct answer is: both spatial and temporal locality

Question 124

Correct

Mark 15.00 out of 15.00

Given a Computer System having a processor with three levels of Caches and Main Memory with the following parameters:

L1 Cache : Hit Rate $hL1$, Hit Time $htL1$ (in clock cycles)

L2 Cache : Hit Rate $hL2$, Hit Time $htL2$ (in clock cycles)

L3 Cache : Hit Rate $hL3$, Hit Time $htL3$ (in clock cycles)

Main Memory : Miss Penalty mp (in clock cycles)

Write the expression for computing the Average Time for accessing the Cache block

Select one:

- ☐ 1. Average Access Time $Tav = htL1 + (1 - htL1)(htL2 + (1 - htL2)(htL3 + (1 - htL3) * mp))$
- ☒ 2. Average Access Time $Tav = htL1 + (1 - hL1)(htL2 + (1 - hL2)(htL3 + (1 - hL3) * mp))$ ✓
- ☐ 3. Average Access Time $Tav = hL1 + (1 - hL1)(hL2 + (1 - hL2)(hL3 + (1 - hL3) * mp))$
- ☐ 4. Average Access Time $Tav = htL1 * (1 - hL1)(htL2 + (1 - hL2)(htL3 + (1 - hL3) * mp))$

The correct answer is: Average Access Time $Tav = htL1 + (1 - hL1)(htL2 + (1 - hL2)(htL3 + (1 - hL3) * mp))$

Question **125**

Correct

Mark 15.00 out of 15.00

Consider a 64-bit microprocessor that has an on-chip 256-KB 16-way Set-Associative mapped cache.

Assume that the cache has a line (block) size of 16 quad words where each quad word is 64-bits (128 bytes).

Where in the cache is the word from memory location CADDECADECAB7BEE mapped?

Select one:

- ☐ 1. TAG=0x32B77B2B7B2AD, SET=0x776, BYTE OFFSET=0x0E
- ☐ 2. TAG=0xCADDECADEC, SET=0xAB7B, BYTE OFFSET=0xEE
- ☐ 3. TAG=0xCADDECADECA, SET=0xB7B, BYTE OFFSET=0xEE
- ☒ 4. TAG=0x32B77B2B7B2AD, SET=0x77, BYTE OFFSET=0x6E ✓

The correct answer is: TAG=0x32B77B2B7B2AD, SET=0x77, BYTE OFFSET=0x6E

Question **126**

Correct

Mark 15.00 out of 15.00

Consider a 32-bit microprocessor that has an on-chip 64KB 4-way set associative cache.

Assume that the cache has a line (block) size of 4 words where each word is 32-bits (4 bytes).

Where in the cache is the word from memory location 79D5C789 mapped?

Select one:

- ☐ 1. TAG=0x1E75, INDEX=0x078, BYTE OFFSET=0x09
- ☒ 2. TAG=0x1E757, INDEX=0x078, BYTE OFFSET=0x09 ✓
- ☐ 3. TAG=0x1E75, INDEX=0x778, BYTE OFFSET=0x09
- ☐ 4. TAG=0x1E757, INDEX=0x778, BYTE OFFSET=0x09

The correct answer is: TAG=0x1E757, INDEX=0x078, BYTE OFFSET=0x09

Question **127**

Correct

Mark 5.00 out of 5.00

If the required word is found in the cache then it is termed as

Select one:

- ☐ 1. Cache Miss
- ☐ 2. Page Fault
- ☐ 3. Page Hit
- ☒ 4. Cache hit ✓

The correct answer is: Cache hit

Question **128**

Correct

Mark 10.00 out of 10.00

What are the main reasons for using Linker apart from the linking process?

Select one:

- ☐ 1. Modularity
- ☐ 2. Time Efficiency
- ☐ 3. Space Efficiency
- ☒ 4. All of the above ✓

The correct answer is: All of the above

Question **129**

Correct

Mark 5.00 out of 5.00

In DRAM, Shortly after the row address is loaded, the column address is applied to the address pins and loaded into the column address latch under the control of the signal.

Select one:

- ☒ 1. Column Address Strobe (CAS) ✓
- ☐ 2. Read/Write (R/W)
- ☐ 3. Chip Select(CS)
- ☐ 4. Row Address Strobe (RAS)

The correct answer is: Column Address Strobe (CAS)

Question 130

Correct

Mark 20.00 out of 20.00

The following program consists of two source files, main.c and swap.c. The main function initializes a two-element array of ints, and then calls the swap function to swap the pair.

```
/* main.c */
void swap();
int buf[2] = {1, 2};
int main() {
    swap();
    return 0;
}

/* swap.c */
extern int buf[];
int *bufp0 = &buf[0];
int *bufp1;
void swap() {
    int temp;
    bufp1 = &buf[1];
    temp = *bufp0;
    *bufp0 = *bufp1;
    *bufp1 = temp;
}
```

For each symbol that is defined or referenced in swap.o, indicate whether or not it will have a symbol table entry in the .symtab section in module swap.o.

If so, indicate the module (swap.o or main.o) that defines the symbol given below, the symbol type (local, global, or extern), and the section (.text, .data, or .bss) it occupies in that module.

Symbol : bufp1

Select one:

- ☐ 1. swap.o .symtab entry? :Yes, Symbol type : global, Module where defined: swap.o, Section:.data
- ☐ 2. swap.o .symtab entry? :Yes, Symbol type : global, Module where defined: swap.o, Section:.text
- ☐ 3. swap.o .symtab entry? :no, Symbol type : -----, Module where defined: -----, Section:-----
- ☒ 4. swap.o .symtab entry? :Yes, Symbol type : global, Module where defined: swap.o, Section:.bss ✓
- ☐ 5. swap.o .symtab entry? :Yes, Symbol type : extern, Module where defined: main.o, Section:.data

The correct answer is: swap.o .symtab entry? :Yes, Symbol type : global, Module where defined: swap.o, Section:.bss

Question **131**

Correct

Mark 15.00 out of 15.00

Consider a 32-bit microprocessor that has an on-chip 64-KB Fully Associative mapped cache.

Assume that the cache has a line (block) size of 4 words where each word is 32-bits (4 bytes).

Where in the cache is the word from memory location A8B7C6D5 mapped?

Select one:

- ☐ 1. TAG=0xA8B7C, BYTE OFFSET=0x6D5
- ☐ 2. TAG=0xA8B7, BYTE OFFSET=0xC6D5
- ☐ 3. TAG=0xA8B7C6, BYTE OFFSET=0xD5
- ☒ 4. TAG=0xA8B7C6D, BYTE OFFSET=0x05 ✓

The correct answer is: TAG=0xA8B7C6D, BYTE OFFSET=0x05

Question **132**

Correct

Mark 10.00 out of 10.00

Linking can be performed at

Select one:

- ☐ 1. Compile time
- ☐ 2. Load time
- ☐ 3. Run time
- ☒ 4. All of the above ✓

The correct answer is: All of the above

Question **133**

Correct

Mark 5.00 out of 5.00

local C variables are stored in either .bss, or .data section

Answer: static ✓

The correct answer is: static

Question **134**

Correct

Mark 5.00 out of 5.00

Static linkers such as the Linux program take as input a collection of relocatable object files and command-line arguments and generate as output a fully linked executable object file that can be loaded and run.

Answer: ld ✓

The correct answer is: ld

Question **135**

Correct

Mark 5.00 out of 5.00

Relocation entries for initialized data are placed in by the assembler.

Select one:

- ☐ 1. .rodata section
- ☐ 2. .rel.text section
- ☒ 3. .rel.data section ✓
- ☐ 4. .bss section

The correct answer is: .rel.data section

Question **136**

Correct

Mark 5.00 out of 5.00

..... is the process of collecting and combining various pieces of code and data into a single file that can be loaded (copied) into memory and executed.

Answer: ✓

The correct answer is: Linking

Question **137**

Correct

Mark 5.00 out of 5.00

A Set Associative Cache has

....., & each set has

Select one:

- ☐ 1. one set, more than one block/line
- ☐ 2. more than one set, one block/line
- ☐ 3. one set, one block/line
- ☒ 4. more than one set, more than one block/line ✓

The correct answer is: more than one set, more than one block/line

Question **138**

Correct

Mark 5.00 out of 5.00

..... write hit policy uses Non-Allocate on write upon a cache miss.

Select one:

- ☐ 1. Non-Allocate on write
- ☒ 2. Write-through ✓
- ☐ 3. Write-back
- ☐ 4. Allocate on write

The correct answer is: Write-through

Question **139**

Correct

Mark 10.00 out of 10.00

..... command is used in linux to disassemble or convert object code to assembly language instructions.

Answer:



The correct answer is: objdump

Question **140**

Correct

Mark 10.00 out of 10.00

Identify all the special pseudo sections that don't have entries in the section header table

Select one:

- ☐ 1. ABS
- ☐ 2. UND
- ☐ 3. COM
- ☒ 4. All of the above ✓

The correct answer is: All of the above

Question **141**

Correct

Mark 5.00 out of 5.00

In a program with good, a memory location that is referenced once is likely to be referenced again multiple times in the near future.

Answer:



The correct answer is: temporal locality

Question **142**

Correct

Mark 10.00 out of 10.00

DLL stands for

Answer:



The correct answer is: Dynamic Link Libraries

Question 143

Correct

Mark 15.00 out of 15.00

Suppose that a 1 MB file consisting of 512-byte logical blocks is stored on a disk drive with the following characteristics:

Parameter	Value
Rotational rate	10,000 RPM
Tavg seek	5 ms
Average # sectors/track	1024
Surfaces	4
Sector size	512 bytes

For the random case below, suppose that a program reads the logical blocks of the file sequentially, one after the other, and that the time to position the head over the first block is Tavg seek + Tavg rotation.

Random case: Estimate the time (in ms) required to read the file if blocks are mapped randomly to disk sectors.

Select one:

- ☐ 1. 8,192ms
- ☐ 2. 6,144ms
- ☒ 3. 16,384ms ✓
- ☐ 4. 10,240ms

The correct answer is: 16,384ms

Question 144

Correct

Mark 5.00 out of 5.00

In a typical ELF(Executable and Linkable Format) relocatable object file, which section/part of the file contains Mapping between line numbers in the original C source program and machine code instructions in the .text section

Select one:

- ☐ 1. .debug section
- ☒ 2. .line section ✓
- ☐ 3. .symtab section
- ☐ 4. .strtab section
- ☐ 5. Section header table
- ☐ 6. .rel.text section
- ☐ 7. .rel.data section

The correct answer is: .line section

Question 145

Correct

Mark 50.00 out of 50.00

Given the following version of the Matrix Multiplication program.

Assume that matrices a, b & c containing elements of the type double are stored in the main memory in row-major order.

The size of the cache block is 8 elements where each element is a double, i.e., size of each element is 8 bytes

Assume that memory system has one level of cache using write-through policy and sum, i, j, k are all mapped to registers

/* ijk */

```
for (i=0; i<80; i++) {  
    for (j=0; j<80; j++) {  
        sum = 0.0;  
        for (k=0; k<80; k++)  
            sum += a[i][k] * b[k][j];  
        c[i][j] = sum;  
    }  
}
```

Determine the total references T in the innermost loop k considering all the k elements accessed where $k = 0, \dots, 79$.

Determine the total number of misses M in the innermost loop k considering all the k elements accessed where $k = 0, \dots, 79$.

Determine the total number of Hits H in the innermost loop k considering all the k elements accessed where $k = 0, \dots, 79$.

Determine the Hit Rate h for the total references in the innermost loop k

Determine the Miss Rate (1-h) for the total references in the innermost loop k

Determine the actual total number of loads L considering each memory read miss as one load

Determine the actual total number of stores S considering each memory write as one store

Select one:

- ☐ 1. T=80, M=10, H=70, $h=0.875$, $(1-h)=0.125$, L=10, S=0
- ☐ 2. T=160, M=90, H=70, $h=0.4375$, $(1-h)=0.5625$, L=90, S=70
- ☐ 3. T=80, M=70, H=10, $h=0.125$, $(1-h)=0.875$, L=70, S=10
- ☒ 4. T=160, M=90, H=70, $h=0.4375$, $(1-h)=0.5625$, L=90, S=0 ✓

The correct answer is: T=160, M=90, H=70, $h=0.4375$, $(1-h)=0.5625$, L=90, S=0

Question 146

Correct

Mark 5.00 out of 5.00

The signals RAS, CAS, Row Address, Column address are generated by the

Select one:

- ☐ 1. Processor
- ☒ 2. DRAM Controller ✓
- ☐ 3. Disk Controller
- ☐ 4. I/O Controller

The correct answer is: DRAM Controller

Question **147**

Correct

Mark 5.00 out of 5.00

Identify the following Compiler Driver tool - Assembler.

Select one:

- ☒ 1. as ✓
- ☐ 2. ld
- ☐ 3. cc1
- ☐ 4. cpp

The correct answer is: as

Question **148**

Correct

Mark 5.00 out of 5.00

State one similarity between SRAM & DRAM semiconductor devices,.....,.....,

Select one:

- ☒ 1. Both are Volatile devices ✓
- ☐ 2. Both are Non-volatile devices
- ☐ 3. Both use capacitors to store the data
- ☐ 4. Both are Electrically Erasable Programmable Read Only Memory devices

The correct answer is: Both are Volatile devices

Question **149**

Correct

Mark 5.00 out of 5.00

The linker blindly performs the relocations using detailed instructions, generated by the assembler called

Select one:

- ☐ 1. assembler directives
- ☐ 2. symbol table entries
- ☒ 3. relocation entries ✓
- ☐ 4. string table entries

The correct answer is: relocation entries

Question **150**

Correct

Mark 15.00 out of 15.00

Consider a 64-bit microprocessor that has an on-chip 128-KB 16-way Set-Associative mapped cache.

Assume that the cache has a line (block) size of 8 quad words where each quad word is 64-bits (8 bytes).

Where in the cache is the word from memory location DAD9FEEDBAC8DEED mapped?

Select one:

- ☒ 1. TAG=0x6D6CFF76DD646, SET=0x7B, BYTE OFFSET=0x2D ✓
- ☐ 2. TAG=0xDAD9FEEDBAC, SET=0x8DE, BYTE OFFSET=0xED
- ☐ 3. TAG=0x6D6CFF76DD64, SET=0x67B, BYTE OFFSET=0x2D
- ☐ 4. TAG=0xDAD9FEEDBA, SET=0xC8DE, BYTE OFFSET=0xED

The correct answer is: TAG=0x6D6CFF76DD646, SET=0x7B, BYTE OFFSET=0x2D

Question **151**

Correct

Mark 15.00 out of 15.00

Consider a 64-bit microprocessor that has an on-chip 256-KB 16-way Set-Associative mapped cache.

Assume that the cache has a line (block) size of 16 quad words where each quad word is 64-bits (128 bytes).

Determine the number of bits in TAG, SET, BYTE OFFSET fields in the memory address.

Select one:

- ☐ 1. TAG=52 bits, SET=8bits, BYTE OFFSET=4 bits
- ☐ 2. TAG=51 bits, SET=9 bits, BYTE OFFSET=4 bits
- ☒ 3. TAG=51 bits, SET=7 bits, BYTE OFFSET=6 bits ✓
- ☐ 4. TAG=50 bits, SET=10 bits, BYTE OFFSET=4 bits

The correct answer is: TAG=51 bits, SET=7 bits, BYTE OFFSET=6 bits

Question **152**

Correct

Mark 10.00 out of 10.00

SRAM STANDS FOR

Answer: ✓

The correct answer is: STATIC RAM

Question 153

Incorrect

Mark 0.00 out of 15.00

Given a Computer System having a processor having clock cycle time 0.33ns with three levels of Caches and Main Memory with the following parameters:

L1 Cache : Hit rate 0.94, Hit Time 2 clock cycles

L2 Cache : Hit rate 0.97, Hit Time 4 clock cycles

L3 Cache : Hit Rate 0.99, Hit Time 8 clock cycles

Main Memory : Miss Penalty 100 clock cycles

Compute the Average Time for accessing the Cache block

Select one:

- ☐ 1. 0.75ns
- ☐ 2. 0.85ns
- ☒ 3. 0.95ns ✖
- ☐ 4. 0.15ns

The correct answer is: 0.75ns

Question 154

Not answered

Marked out of 5.00

In a typical ELF(Executable and Linkable Format) relocatable object file, which section/part of the file contains Initialized global and initialized static C variables. Local C variables are maintained at run time on the stack and do not appear in either the .data or .bss sections.

Select one:

- ☐ 1. .bss section
- ☐ 2. .rodata section
- ☐ 3. Section header table
- ☐ 4. .text section
- ☐ 5. .data section
- ☐ 6. ELF header
- ☐ 7. Segment header table

The correct answer is: .data section

Question 155

Correct

Mark 5.00 out of 5.00

In a typical ELF(Executable and Linkable Format) relocatable object file, which section/part of the file contains Offsets and sizes of each section.

Select one:

- ☐ 1. Segment header table
- ☐ 2. .rodata section
- ☐ 3. .bss section
- ☐ 4. .data section
- ☒ 5. Section header table ✔
- ☐ 6. .text section
- ☐ 7. ELF header

The correct answer is: Section header table

Question **156**

Correct

Mark 5.00 out of 5.00

.....,....., occurs because the cache is initially empty.

Select one:

- ☐ 1. TBL miss
- ☐ 2. Conflict miss
- ☐ 3. Capacity miss
- ☒ 4. Cold(Compulsory) miss ✓

The correct answer is: Cold(Compulsory) miss

Question **157**

Correct

Mark 5.00 out of 5.00

In order to store a bit 0 in static RAM cell, the bit lines b & b' connected to the pass transistors must be given logic levels :

Select one:

- ☐ 1. 1, 0
- ☐ 2. 0, 0
- ☐ 3. 1, 1
- ☒ 4. 0, 1 ✓

The correct answers are: 0, 1, 1, 0

Question **158**

Correct

Mark 5.00 out of 5.00

Identify the following Compiler Driver tool - Static linker.

Select one:

- ☐ 1. cc1
- ☐ 2. as
- ☒ 3. ld ✓
- ☐ 4. cpp

The correct answer is: ld

Question **159**

Correct

Mark 10.00 out of 10.00

..... SDRAM is an enhancement of SDRAM that doubles the speed of the DRAM by using both clock edges as control signals.

Answer: ✓

The correct answer is: DDR

Question 160

Incorrect

Mark 0.00 out of 15.00

Given a Seagate Cheetah 15K.4 Hard disk drive containing 4 platters having a zone map in the table shown below.

Determine the number of spare cylinders in the Zone 8.

Zone Number	Sectors per track	Cylinders per zone	Logical Blocks per zone
0	864	3201	22,076,928
1	844	3200	21,559,136
2	816	3400	22,149,504
3	806	3100	19,943,664
4	795	3100	19,671,480
5	768	3400	20,852,736
6	760	3450	21,159,936
7	725	3650	21,135,200
8	704	3700	20,804,608
9	672	3700	19,858,944
10	640	3700	18,913,280
11	603	3700	17,819,856
12	576	3707	17,054,208
13	528	3060	12,900,096

Select one:

- ☒ 1. 7 ✖
- ☐ 2. 8
- ☐ 3. 6
- ☐ 4. 9

The correct answer is: 6

Question 161

Correct

Mark 10.00 out of 10.00

SDRAM stands for

Answer: Synchronous DRAM ✓

The correct answer is: Synchronous DRAM

Question 162

Correct

Mark 5.00 out of 5.00

Symbol table is built by, using symbols exported by the compiler into the assembly-language .s file.

Answer: assembler ✓

The correct answer is: assembler

Question 163

Not answered

Marked out of
50.00

Given the following version of the Matrix Multiplication program.

Assume that matrices a, b & c containing elements of the type double are stored in the main memory in row-major order.

The size of the cache block is 8 elements where each element is a double, i.e., size of each element is 8 bytes

Assume that memory system has one level of cache using write-through policy and r, i, j, k are all mapped to registers

```
/* ikj */
for (i=0; i<80; i++) {
    for (k=0; k<80; k++) {
        r = a[i][k];
        for (j=0; j<80; j++)
            c[i][j] += r * b[k][j];
    }
}
```

Determine the total references T in the innermost loop j considering all the j elements accessed where j =0, ...,79.

Determine the total number of misses M in the innermost loop j considering all the j elements accessed where j =0, ...,79.

Determine the total number of Hits H in the innermost loop j considering all the j elements accessed where j =0, ...,79.

Determine the Hit Rate h for the total references in the innermost loop j

Determine the Miss Rate (1-h) for the total references in the innermost loop j

Determine the actual total number of loads L considering each memory read miss as one load

Determine the actual total number of stores S considering each memory write as one store

Select one:

- ☐ 1. T=160, M=20, H=140, h=0.875, (1-h)=0.125, L=20, S=160
- ☐ 2. T=160, M=20, H=140, h=0.875, (1-h)=0.125, L=20, S=80
- ☐ 3. T=80, M=10, H=70, h=0.875, (1-h)=0.125, L=10, S=0
- ☐ 4. T=80, M=70, H=10, h=0.125, (1-h)=0.875, L=70, S=80

The correct answer is: T=160, M=20, H=140, h=0.875, (1-h)=0.125, L=20, S=80

Question 164

Not answered

Marked out of
5.00

The purpose of step of the linker is to merge the similar sections in the object files and relocate these merged sections by associating a memory location with each symbol definition, and then modifying all of the references to those symbols so that they point to this memory location.

Answer: ✖

The correct answer is: Relocation

Question **165**

Not answered
Marked out of
5.00

In a typical ELF(Executable and Linkable Format) relocatable object file, which section/part of the file contains Uninitialized global and uninitialized static C variables.

Select one:

- ☐ 1. .bss section
- ☐ 2. Segment header table
- ☐ 3. .rodata section
- ☐ 4. Section header table
- ☐ 5. ELF header
- ☐ 6. .data section
- ☐ 7. .text section

The correct answer is: .bss section

Question **166**

Not answered
Marked out of
5.00

Examples of Linker Global symbols are

Select one:

- ☐ 1. C functions and global variables declared with the extern attribute
- ☐ 2. Non-static C functions and non-static global variables
- ☐ 3. C functions and global variables defined with the static attribute
- ☐ 4. Local program variables

The correct answer is: Non-static C functions and non-static global variables

Question **167**

Not answered
Marked out of
5.00

In order to store a bit 1 in static RAM cell, the bit lines b & b' connected to the pass transistors must be given logic levels

Select one:

- ☐ 1. 0, 0
- ☐ 2. 0, 1
- ☐ 3. 1, 1
- ☐ 4. 1, 0

The correct answer is: 1, 0

Question **168**

Not answered
Marked out of
5.00

Provide the name of the executable object file generated by the following command :

`$gcc sum.c`

Answer: 

The correct answer is: a.out

Question **169**

Not answered

Marked out of 5.00

In Cache memory, each Word is a collection of one or more

Answer: ✖

The correct answer is: bytes

Question **170**

Not answered

Marked out of 5.00

..... library inter-positioning requires access only to the executable object file.

Select one:

- ☐ 1. Run-time
- ☐ 2. Link-time
- ☐ 3. Compile time
- ☐ 4. Not a single

The correct answer is: Run-time

Question **171**

Not answered

Marked out of 10.00

To reduce the number of pins needed for external connections, the row and column addresses in DRAM are.....

Answer: ✖

The correct answer is: multiplexed

Question **172**

Not answered

Marked out of 20.00

Given the following statements in an relocatable object file generated by an assembler:

```
e: e8 00 00 00 00    callq sum           # sum()
      f : R_X86_64_PC32 sum -0x4      # Relocation
entry
```

Determine the following fields in the Relocation entry : r.offset, r.type, r.symbol, r.addend

Select one:

- ☐ 1. r.offset = 0xf, r.type = R_X86_64_PC32, r.symbol = sum, r.addend = - 0x4
- ☐ 2. r.offset = 0xf, r.type = R_X86_64_PC32, r.symbol = - 0x4, r.addend = sum
- ☐ 3. r.offset = -0x4, r.type = sum, r.symbol = R_X86_64_PC32, r.addend = 0xf
- ☐ 4. r.offset = -0x4, r.type = R_X86_64_PC32, r.symbol = sum, r.addend = 0xf

The correct answer is: r.offset = 0xf, r.type = R_X86_64_PC32, r.symbol = sum, r.addend = - 0x4

Question **173**

Not answered

Marked out of 5.00

Dynamic linker tool used on linux system is

Select one:

- ☐ 1. execve
- ☐ 2. ld-linux
- ☐ 3. ld
- ☐ 4. syscall

The correct answer is: ld-linux

Question **174**

Not answered

Marked out of 5.00

Fraction of memory references not found in cache (misses / accesses) is termed as.....

Select one:

- ☐ 1. Miss Rate
- ☐ 2. Miss Penalty
- ☐ 3. Hit Rate
- ☐ 4. Hit Time

The correct answer is: Miss Rate

Question **175**

Not answered

Marked out of 10.00

MOSFET STANDS FOR

Answer:



The correct answer is: METAL OXIDE SEMICONDUCTOR FIELD EFFECT TRANSISTOR

Question **176**

Not answered

Marked out of 5.00

In a typical ELF(Executable and Linkable Format) relocatable object file, which section/part of the file contains machine code of the compiled program.

Select one:

- ☐ 1. Segment header table
- ☐ 2. .rodata section
- ☐ 3. ELF header
- ☐ 4. .bss section
- ☐ 5. Section header table
- ☐ 6. .data section
- ☐ 7. .text section

The correct answer is: .text section

Question **177**

Not answered

Marked out of
5.00

Identify the following Compiler Driver tool - Compiler.

Select one:

- ☐ 1. cpp
- ☐ 2. as
- ☐ 3. cc1
- ☐ 4. ld

The correct answer is: cc1

Question **178**

Not answered

Marked out of
15.00

Consider a 32-bit microprocessor that has an on-chip 64KB Fully Associative mapped cache.

Assume that the cache has a line (block) size of 4 words where each word is 32-bits (4 bytes).

Determine the number of bits in TAG & BYTE OFFSET fields in the memory address.

Select one:

- ☐ 1. TAG=28 bits, BYTE OFFSET= 4 bits
- ☐ 2. TAG=22 bits, BYTE OFFSET=10 bits
- ☐ 3. TAG=20 bits, BYTE OFFSET=12 bits
- ☐ 4. TAG=24 bits, BYTE OFFSET= 8 bits

The correct answer is: TAG=28 bits, BYTE OFFSET= 4 bits